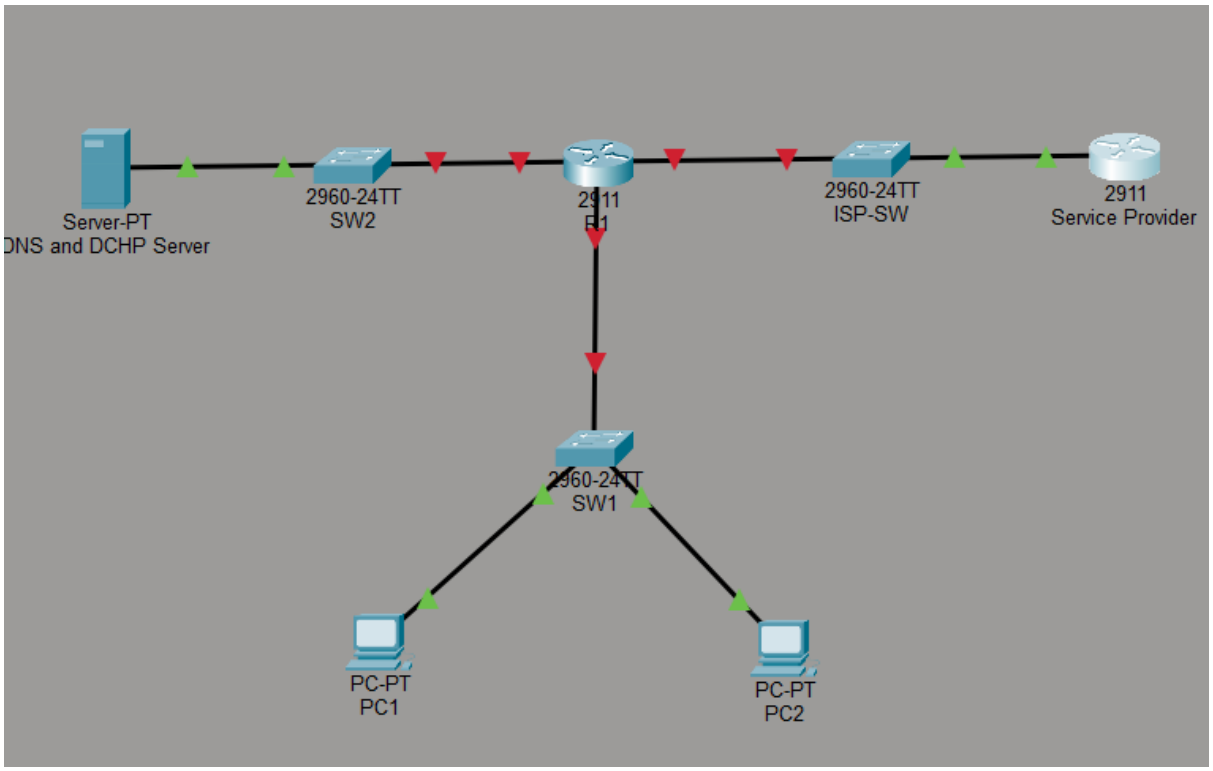
Network Topology



Service Provider IP

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.100.1   YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

Copy Paste

ISP DHCP Pool

```
Building configuration...
[OK]
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.100.1 192.168.100.10
Router(config)#ip dhcp pool ISP_POOL
Router(dhcp-config)#network 192.168.100.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.100.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#show ip dhcp pool

Pool ISP_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 1
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.100.1        192.168.100.1    - 192.168.100.254   0    / 1     / 254
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

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R1 DHCP Client

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address dhcp
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES DHCP   up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

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R1 LAN Interface

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/2
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.100.11  YES DHCP   up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     192.168.10.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

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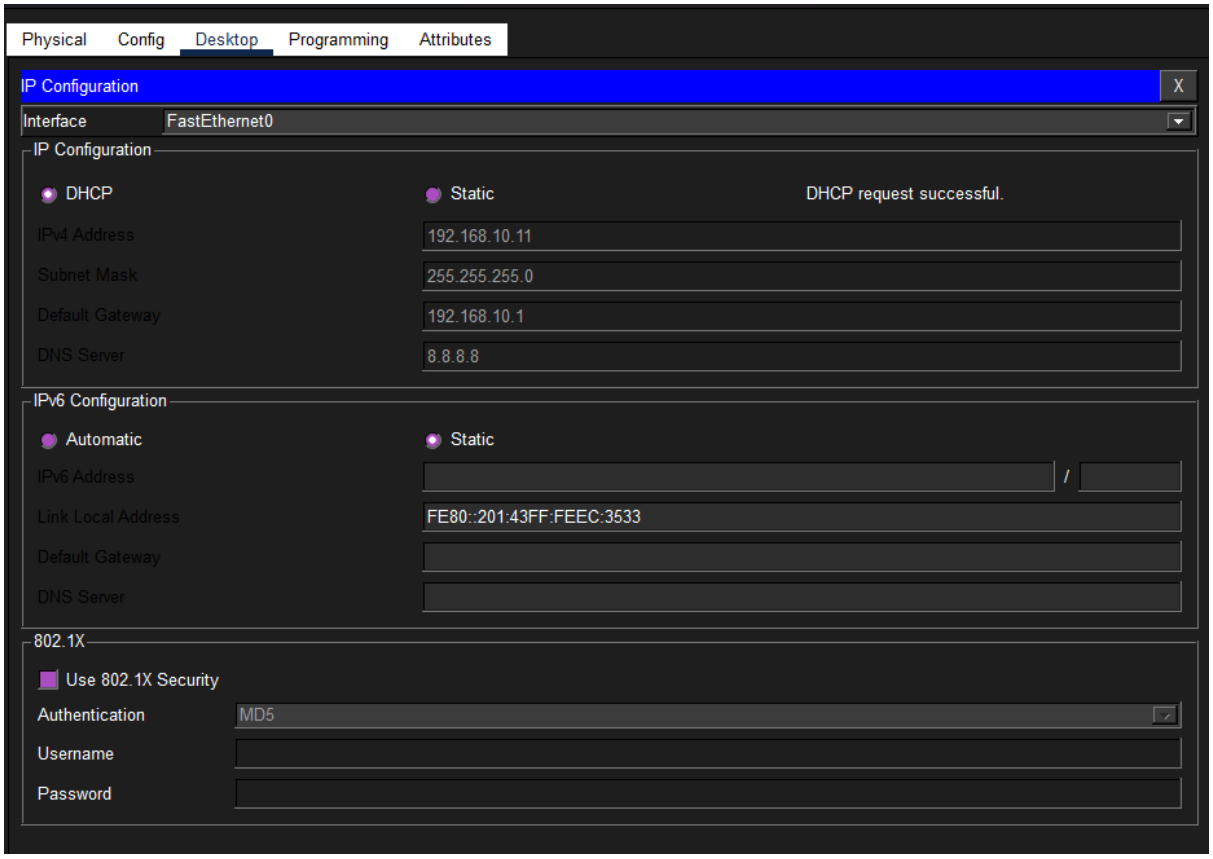
R1 DHCP Pool

```
Pool LAN_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 1
 Pending event                  : none

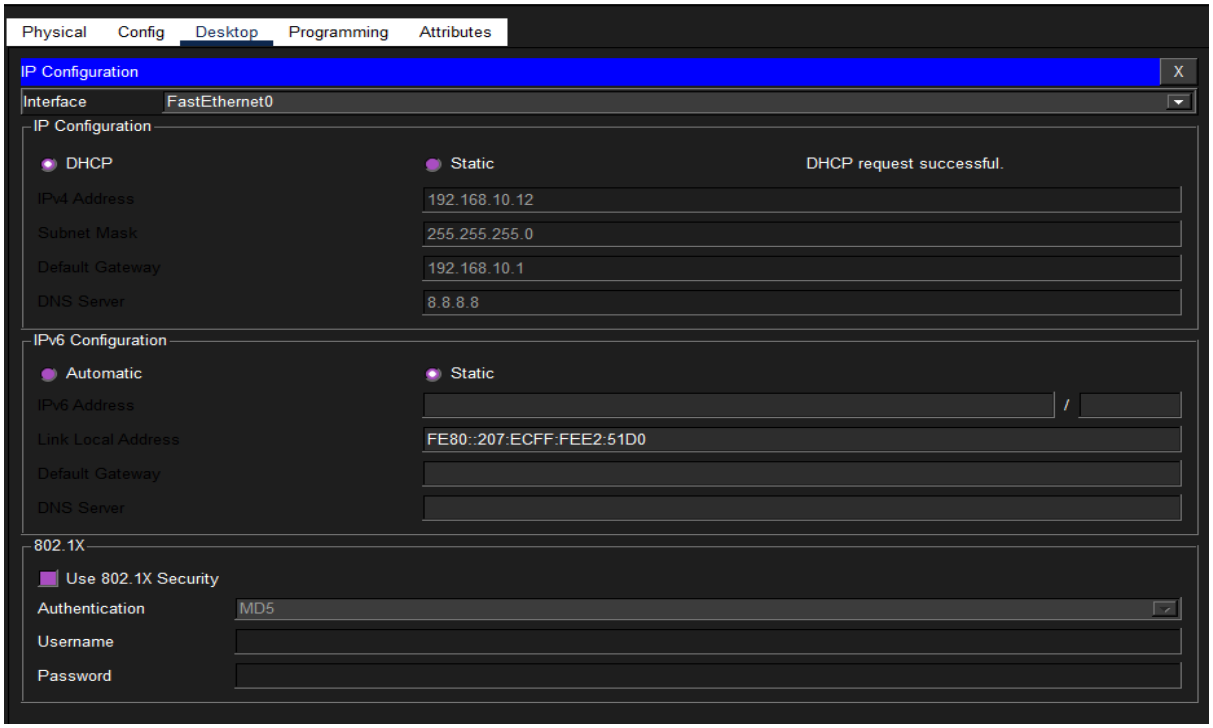
 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    0    / 1     / 254
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

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PC1 DHCP Configuration



PC2 DHCP Configuration

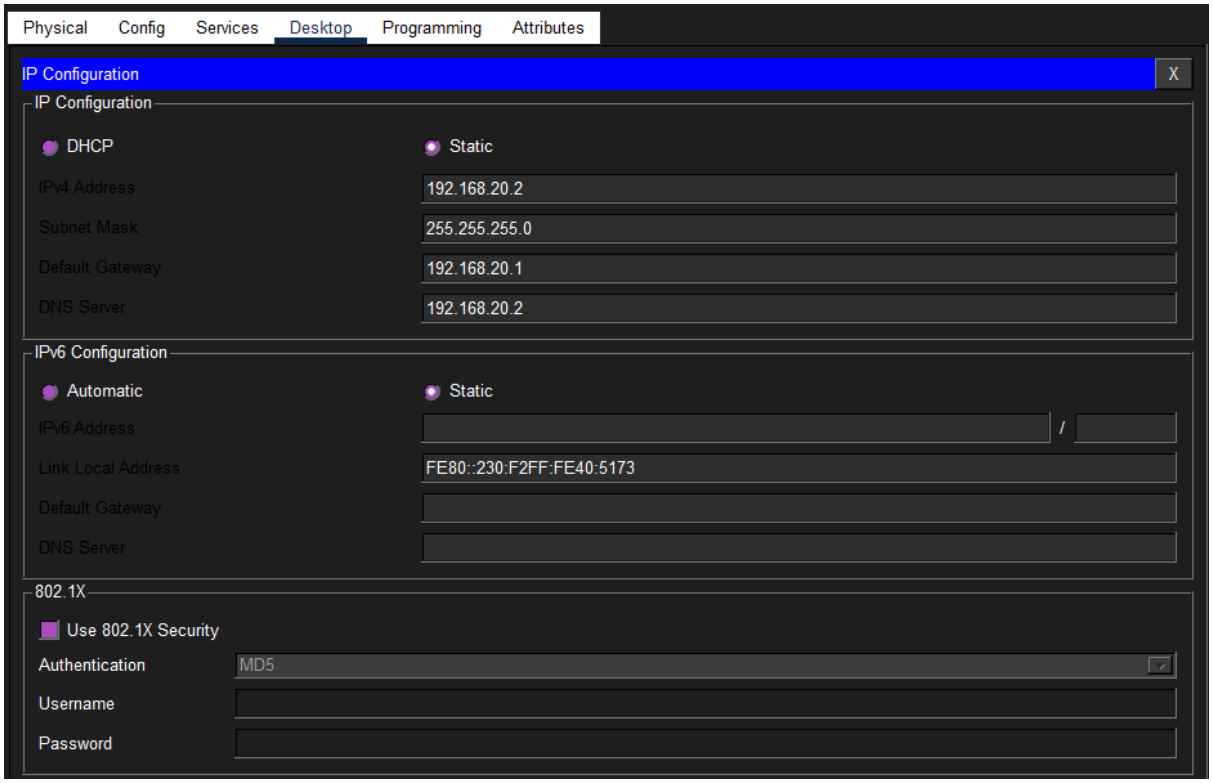


DHCP Bindings

```
Building configuration...
[OK]
Router#enable
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.11    0001.43EC.3533           --                     Automatic
192.168.10.12    0007.ECE2.51D0           --                     Automatic
Router#
```

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External Server IP



Server Network Interface

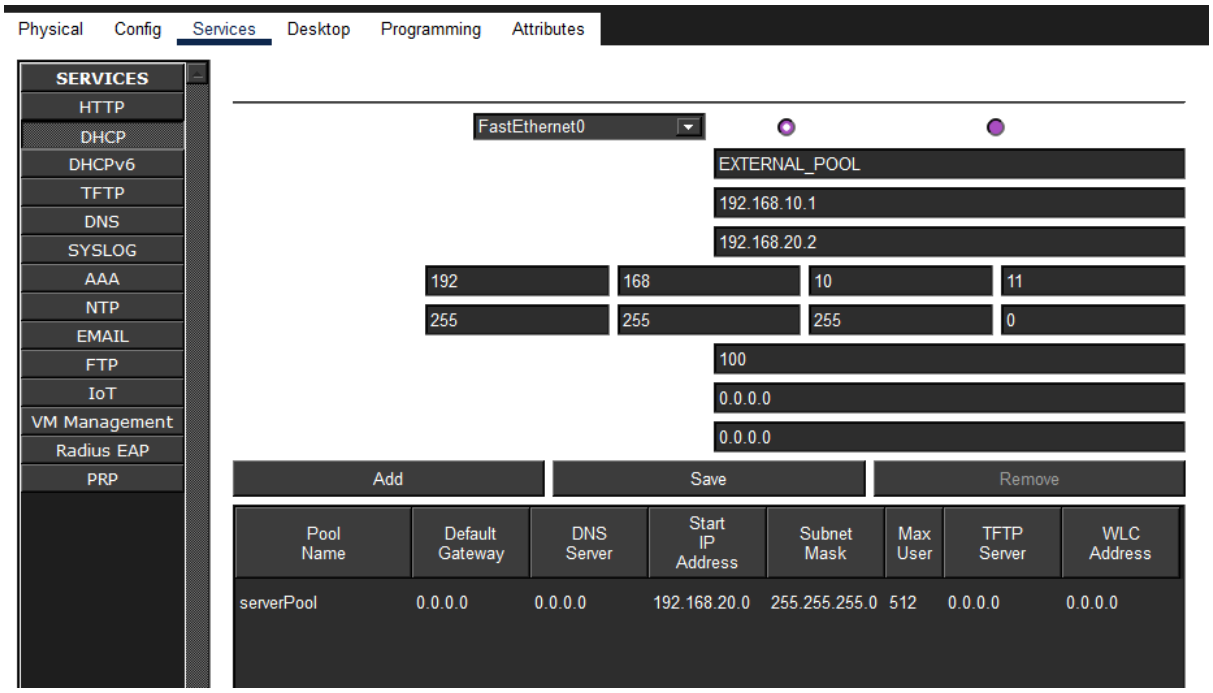
```
Router(config-if)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.100.11  YES DHCP   up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     192.168.10.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

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External DHCP Pool



DHCP Relay Configuration

```
Building configuration...
[OK]
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/2
Router(config-if)#ip helper-address 192.168.20.2
Router(config-if)#exit
Router(config)#end
Router#show running-config interface gigabitethernet0/2
                            ^
% Invalid input detected at '^' marker.

Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

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External DHCP on PC1

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /releaseipconfig /renewipconfig /all
Invalid Command.

C:\>ipconfig /release

   IP Address......................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: 0.0.0.0
   DNS Server......................: 0.0.0.0

C:\>ipconfig /renew

   IP Address......................: 192.168.10.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.10.1
   DNS Server......................: 192.168.20.2

C:\>ipconfig /all

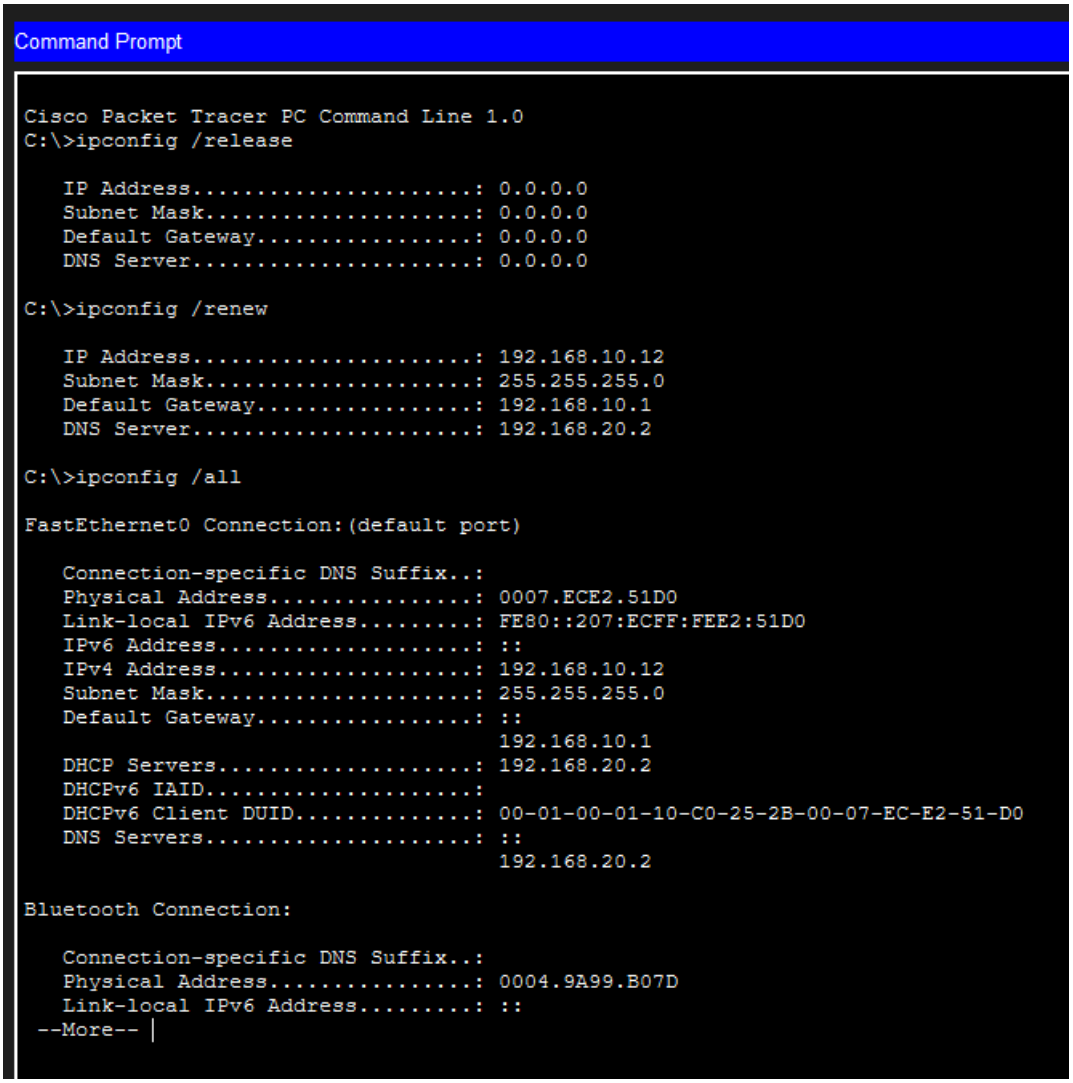
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.43EC.3533
   Link-local IPv6 Address.........: FE80::201:43FF:FEEC:3533
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1
   DHCP Servers....................: 192.168.20.2
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-28-E4-E3-2A-00-01-43-EC-35-33
   DNS Servers.....................: ::
                                     192.168.20.2

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 0004.9A1C.ED64
   Link-local IPv6 Address.........: ::
 --More--
```

PC2 External DHCP



DHCP Reflection

A router acting as a DHCP server gives IP addresses directly to devices on the network, an external DHCP server is a separate device that manages IP addresses and offers more control and flexibility, a router DHCP server is suitable for smaller networks, while an external DHCP server is better for larger networks with more devices.